Supplementary Information

Certified discovery: statistically valid materials screening claims from a small DFT verification budget

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This file contains Supplementary Notes 1–4 and Supplementary Tables 1–3.

Supplementary Note 1: Asymptotics of the plug-in coefficient

Main-text Theorem 1, Corollary 1 and Propositions 2–3 are exact finite-population statements and are proved in full in the main text. Main-text Proposition 1 is asymptotic; it coincides with the classical theory of the design-based regression estimator under simple random sampling without replacement (Neyman 1934; Cochran 1977, ch. 7). This note writes out that argument specialized to our bounded setting, with every assumption stated.

Setting and assumptions. A finite population $\{(y_i, f_i)\}_{i=1}^N$ with $y_i \in \{0,1\}$, $f_i \in [0,1]$, mean $\pi = \bar{y}$, prediction mean $\bar{f}$, variances S_y^2, S_f^2 and covariance S_{yf} (all $\frac{1}{N-1}$-normalized); a simple random sample without replacement $\mathscr{L}$ of size n; the estimator $\hat{\pi}_{\hat{\lambda}}$ of main-text equation (1) with the sample regression slope $\hat{\lambda} = \widehat{S}_{yf}/\widehat{S}_f^2$ of main-text equation (2), and $\hat{\lambda} := 0$ if $\widehat{S}_f^2 = 0$. The oracle coefficient is $\lambda^\star = S_{yf}/S_f^2$ (main-text Theorem 1(iii)). We consider a sequence of populations with $n \to \infty$, $N - n \to \infty$, $\liminf S_y^2 > 0$, $\liminf S_f^2 > 0$ and $\limsup \rho^2 < 1$; the last two conditions together are equivalent to $\liminf S_r^2(\lambda^\star) > 0$, since $S_r^2(\lambda^\star) = (1-\rho^2)S_y^2$, and rule out the degenerate case in which the rectifier population has vanishing variance. For binary labels $\liminf S_y^2 > 0$ says the pools are not asymptotically pure, i.e. π stays bounded away from 0 and 1, a condition every campaign can check on its own labeled sample.

Step 1 (central limit theorem at fixed λ). By main-text Theorem 1, $\hat{\pi}_\lambda$ is unbiased with exact variance $\frac{1}{n}\big(1 - \frac{n}{N}\big)S_r^2(\lambda)$. The rectifier population $\{y_i - \lambda f_i\}$ is uniformly bounded by $1 + |\lambda|$, so with $n \to \infty$, $N - n \to \infty$ and $\liminf S_r^2(\lambda) > 0$ the Hájek–Lindeberg negligibility condition $\max_i (r_i - \bar{r})^2 / \big(n S_r^2\big) \to 0$ holds automatically, and the Hájek finite-population central limit theorem gives asymptotic normality of the standardized rectifier mean; at $\lambda = \lambda^\star$ the required nondegeneracy is exactly $\liminf S_y^2 > 0$ together with $\limsup \rho^2 < 1$.

Step 2 (estimated λ). Writing $\hat{\pi}_\lambda = \bar{y}_{\mathscr{L}} + \lambda(\bar{f} - \bar{f}_{\mathscr{L}})$ shows that the dependence on λ is affine, so

$$\hat{\pi}_{\hat{\lambda}} - \hat{\pi}_{\lambda^\star} = (\hat{\lambda} - \lambda^\star)\big(\bar{f} - \bar{f}_{\mathscr{L}}\big)$$

holds as an exact algebraic identity. Sample means, variances and covariances under simple random sampling without replacement of bounded populations are $\sqrt{n}$-consistent for their population counterparts (their design variances are $O(1/n)$ by the same without-replacement formula, and all population moments are bounded because y and f are), so $\widehat{S}_f^2 \to S_f^2 > 0$ in probability and $\hat{\lambda} - \lambda^\star = O_p(n^{-1/2})$; likewise $\bar{f} - \bar{f}_{\mathscr{L}} = O_p(n^{-1/2})$ by Step 1 applied to the population $\{f_i\}$ (or directly from its design variance). The product is therefore $O_p(n^{-1})$, negligible against the $O_p(n^{-1/2})$ leading term, so $\hat{\pi}_{\hat{\lambda}}$ has the same limiting distribution as $\hat{\pi}_{\lambda^\star}$.

Step 3 (variance estimation; stated exactly at fixed λ as main-text Lemma 1). The plug-in $\widehat{\mathrm{se}}^2$ evaluates the exact variance formula of Theorem 1(ii) at $\hat{\lambda}$ with sample moments; consistency of those moments and continuity of $S_r^2(\cdot)$ give $\widehat{\mathrm{se}}^2/\mathrm{Var}(\hat{\pi}_{\lambda^\star}) \to 1$ in probability, and Slutsky's theorem yields asymptotic coverage $1-\alpha$ for the studentized interval. The implementation additionally retains a $\hat{\lambda}^2 S_f^2/N$ term even though the pool prediction average is deterministic given S; this term is nonnegative, so it can only enlarge $\widehat{\mathrm{se}}$ and cannot reduce coverage below the nominal level, which is why the main text describes the interval as attaining coverage at least $1-\alpha$ asymptotically.

Step 4 (never-wider property). At $\lambda = 0$ the estimator reduces exactly to the classical label-only estimator. Since $\hat{\lambda} \to \lambda^\star$ in probability and the asymptotic variance $S_r^2(\lambda)$ is continuous in λ with $S_r^2(\lambda^\star) = (1-\rho^2)S_y^2 \leq S_y^2 = S_r^2(0)$ (main-text Theorem 1(iii)), the tuned interval's asymptotic width is bounded by the classical width, with equality if and only if $S_{yf} = 0$.

Empirical coverage at every configuration used in the paper is reported in Supplementary Table 1 as a complementary check.

Supplementary Note 2: Neyman allocation for rectifier variances (classical)

This note records, for completeness and attribution, the classical derivation behind the stratified budget rule of the main text; nothing here is novel (Neyman 1934; Cochran 1977, ch. 5).

With strata of sizes N_k, weights $w_k = N_k/N$, per-stratum rectifier variances $S_{r,k}^2$ and allocations n_k summing to n, the stratified certification estimator has variance

$$V(\{n_k\}) = \sum_k w_k^2 S_{r,k}^2 \Big(\frac{1}{n_k} - \frac{1}{N_k}\Big).$$

Minimizing $\sum_k w_k^2 S_{r,k}^2/n_k$ subject to $\sum_k n_k = n$ by Cauchy–Schwarz (or a Lagrange multiplier) gives $n_k \propto w_k S_{r,k} \propto N_k S_{r,k}$, Neyman's rule; the $-1/N_k$ terms do not affect the minimizer. Equal per-stratum variances reduce it to proportional allocation.

Two-stage pilot. $S_{r,k}$ is unknown before labeling. The toolkit spends a small uniform pilot (default 100 labels) to estimate each $S_{r,k}$ at the working coefficient, allocates the remaining budget by the plug-in Neyman rule with a floor of two labels per stratum, and computes the final intervals on the post-pilot samples only, so the pilot does not contaminate the design. The efficiency of the plug-in rule relative to uniform allocation is quantified in Supplementary Table 2; a formal regret bound relative to oracle allocation is left as future work and is not claimed.

Supplementary Note 3: Small-sample and rare-positive regime

When n is small or true positives are rare inside the pool, the normal approximation for the rectifier degrades. The toolkit then falls back to $\lambda = 0$, i.e. the classical label-only estimator, with a Wilson score interval on the labeled proportion; the switch is triggered when the labeled count of positives or negatives falls below 10, and the fallback omits the finite-population correction, so it is conservative by construction. The cost of the fallback is the loss of the prediction-powered width reduction in exactly the regime where predictions are hardest to calibrate, which we regard as the appropriate trade-off. Empirical coverage across models, thresholds, pools and budgets is reported in Supplementary Table 1; the fallback was not triggered in any configuration of the main experiments.

Supplementary Note 4: What the protocol does not certify

The protocol certifies statements of the form "the flagged pool contains $C \in [\ell, u]$ materials with $E_{\text{hull}} \leq \tau$ under the stated DFT settings, at 95% confidence." It does not certify novelty relative to prior databases, synthesizability, functional performance, or any property not encoded in the oracle. Each of those requires its own oracle and, given one, the same machinery applies.

Supplementary Table 1 Empirical coverage of nominal 95% certified intervals against census truth. One row per model, one column per (pool, threshold) at verification budget $n = 500$; entries are the fraction of 1,000 Monte Carlo replicates whose certified interval contains the census value, on the full WBM pool and the redundancy-controlled unique-prototype pool. All ninety values lie in $[0.924, 0.967]$ (median 0.945); eighty-six are within three Monte Carlo standard errors of nominal, and the four below that band (all at the loosest threshold $\tau = 0.10$) occur at rates consistent with Monte Carlo fluctuation at nominal coverage. Classical (label-only) intervals behave similarly: across the budget sweep of main-text Fig. 3a (budgets 50–3,200) their coverage ranges over $[0.906, 0.976]$, with the low end at the smallest budgets where the normal approximation is weakest. Values below the three-Monte-Carlo-standard-error floor of 0.929 are printed in bold vermillion.

	Full pool			Unique prototypes		
Model	$\tau = 0$	0.05	0.10	$\tau = 0$	0.05	0.10
EquiformerV3+DeNS-OAM	0.931	0.932	0.935	0.945	0.937	**0.926**
eSEN-30M-OAM	0.946	0.948	0.939	0.938	0.936	0.949
PET-OAM-XL	0.944	0.936	**0.928**	0.956	0.940	0.949
MatRIS-10M-OAM	0.955	0.945	0.945	0.952	0.946	0.949
ORB v3	0.956	0.942	0.946	0.945	0.944	0.946
MatterSim v1 5M	0.951	0.933	0.934	0.939	0.933	0.960
SevenNet-MF-ompa	0.955	0.941	0.945	0.943	0.951	0.947
GRACE-2L-OAM	0.953	0.951	0.940	0.939	0.940	**0.925**
DPA3-v1-OpenLAM	0.942	0.938	0.947	0.945	0.940	0.942
MACE-MPA-0	0.952	0.935	**0.924**	0.934	0.953	0.944
CHGNet	0.953	0.952	0.948	0.952	0.942	0.942
M3GNet	0.952	0.944	0.967	0.954	0.937	0.952
ALIGNN	0.946	0.946	0.947	0.942	0.946	0.953
MEGNet	0.963	0.948	0.945	0.955	0.954	0.941
CGCNN	0.934	0.950	0.949	0.953	0.940	0.940

Supplementary Table 2 Stratified versus uniform allocation at a fixed verification budget. Interval-width ratios below one favor stratification; the 100-label pilot is drawn uniformly and excluded from the final intervals (Supplementary Note 2); 200 replicates. Two stratification schemes are shown, model-confidence tertiles and composition arity, for three models spanning the accuracy range. Gains are modest at best in this setting: the largest reduction is 8% (GRACE-2L-OAM, confidence tertiles), and composition-arity strata offer no benefit.

Stratification scheme	Model	Budget n	Pilot size	Width ratio[a]
confidence tertiles	EquiformerV3+DeNS-OAM	500	100	0.994
confidence tertiles	GRACE-2L-OAM	500	100	0.920
confidence tertiles	CGCNN	500	100	0.994
composition arity	EquiformerV3+DeNS-OAM	500	100	1.006
composition arity	GRACE-2L-OAM	500	100	1.061
composition arity	CGCNN	500	100	0.995

[a]Mean certified-interval width under Neyman allocation divided by the width under uniform allocation at the same total budget.

Supplementary Table 3 Robustness of certified precision to threshold perturbation and to margin-valued predictions (representative model: EquiformerV3+DeNS-OAM, the lowest-MAE model of the run; full WBM pool). Perturbing the stability threshold by ± 10 meV/atom redefines the flagged pool and therefore the estimand itself; certification is threshold-agnostic and tracks the redefined pool. Substituting continuous stability margins for binary calls tightens the certified interval at the same verification budget (mean width 0.049 versus 0.051 at the unperturbed threshold). Each row aggregates 200 replicate campaigns at a verification budget of $n = 500$ labels; the coverage column is the fraction of replicates whose interval contains the census truth.

Prediction type	Threshold	Pool size N	Mean width	Coverage
margins	$\tau - 0.01$	32,876	0.045	0.920
margins	$\tau + 0.00$	43,437	0.049	0.945
margins	$\tau + 0.01$	59,130	0.042	0.900
binary calls	$\tau + 0.00$	43,437	0.051	0.945